Information System: QGIS South Wales Crime Maps

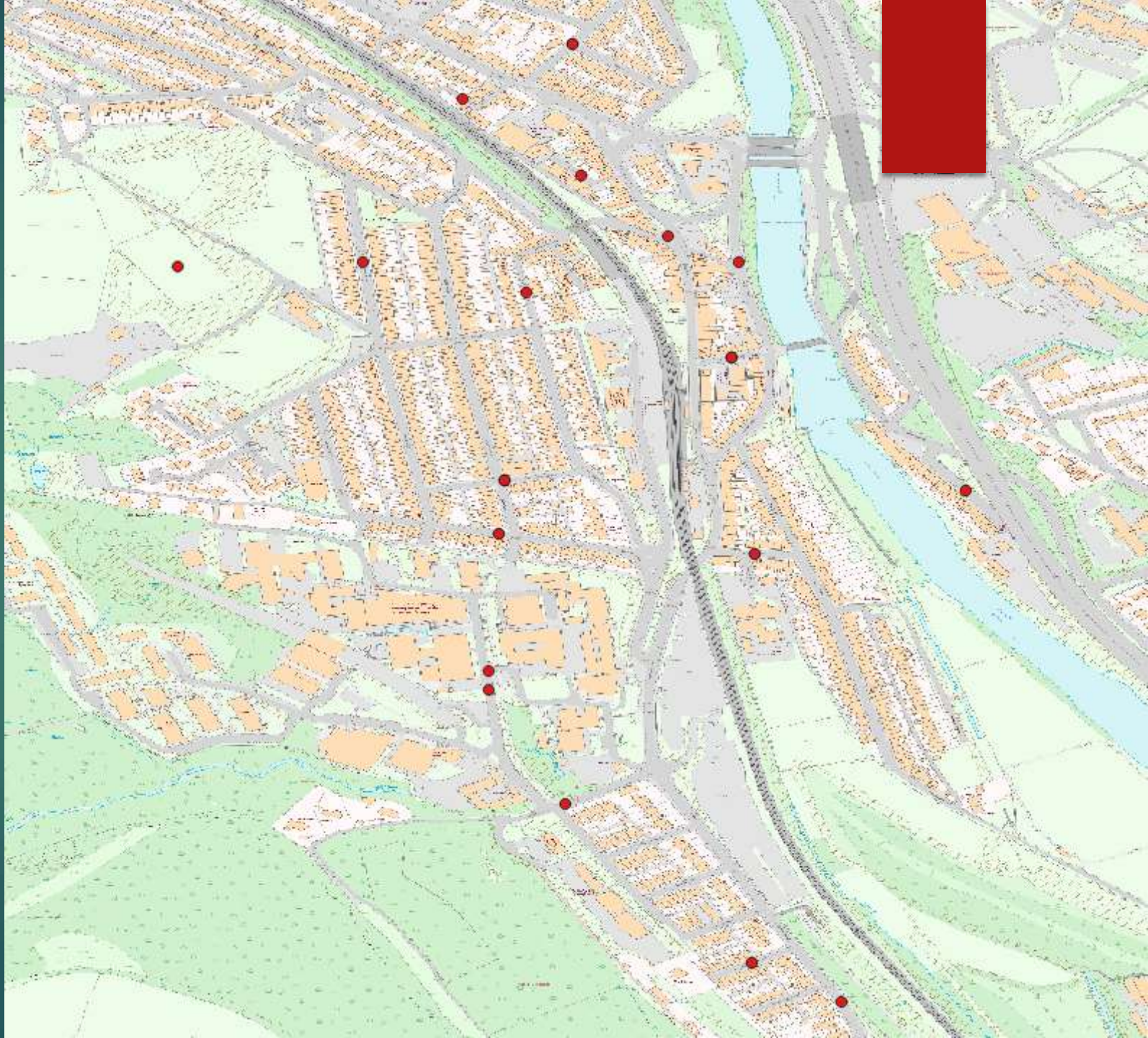
18024092 – FEALY

Map data

All crime data required for the following maps was gathered from the Police UK website (Police.uk, 2019). The follow maps focus on the months of January 2018 and January 2019, this data is categorised monthly on the Police UK website.

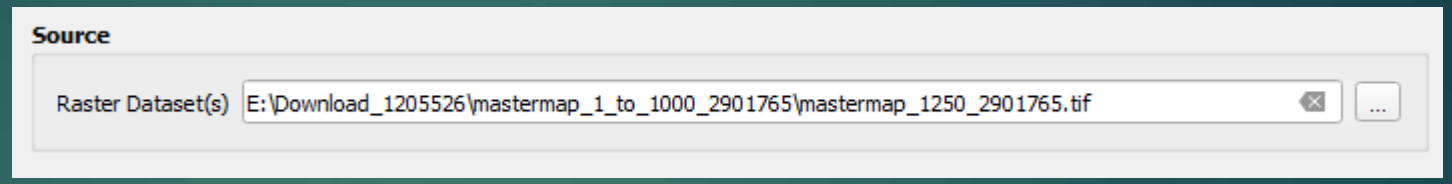
The raster layer used in the follow maps was acquired from Digimap (Digimap.edina.ac.uk, 2019). Through Digimap I selected the treforest region and downloaded the master map (1: 1000) for the selected region.

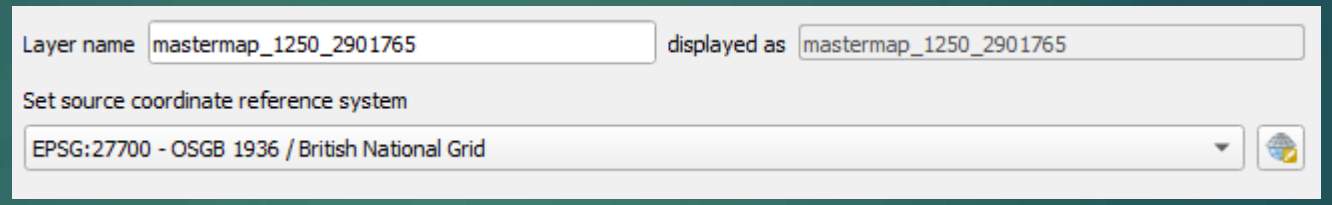
All Crimes points during Jan 2018



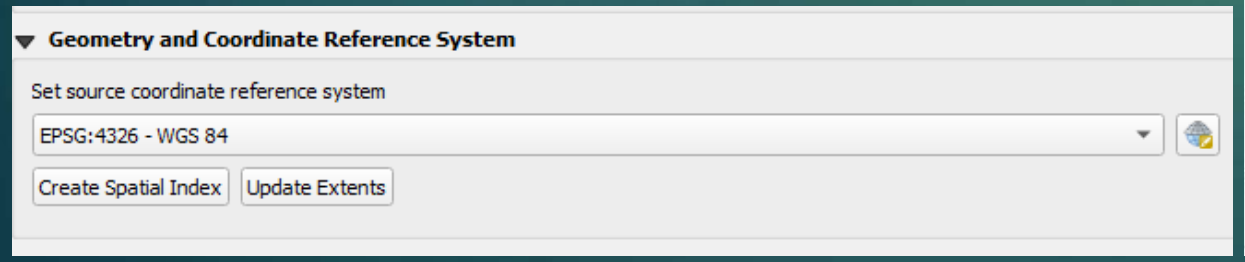
Steps taken to produce map 1

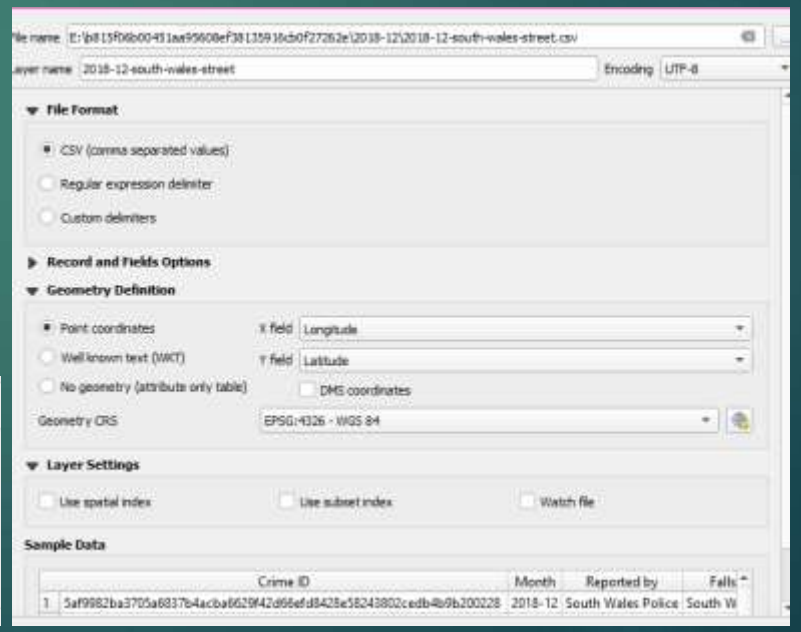
To generate map one I need to add a new raster layer which adds my digimap TIF file to the project. Using the EPSG:27700 positioning settings.





To add crime data to this map, I created a delimited text layer. Which allows me to add my crime data. This layers positioning layer uses EPSG:4326.



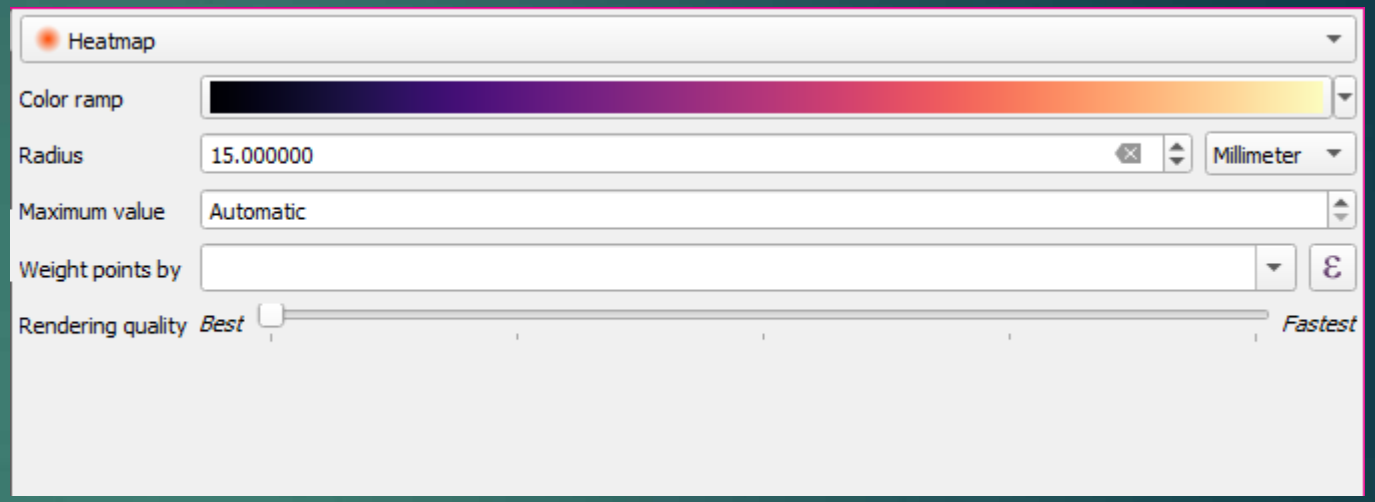


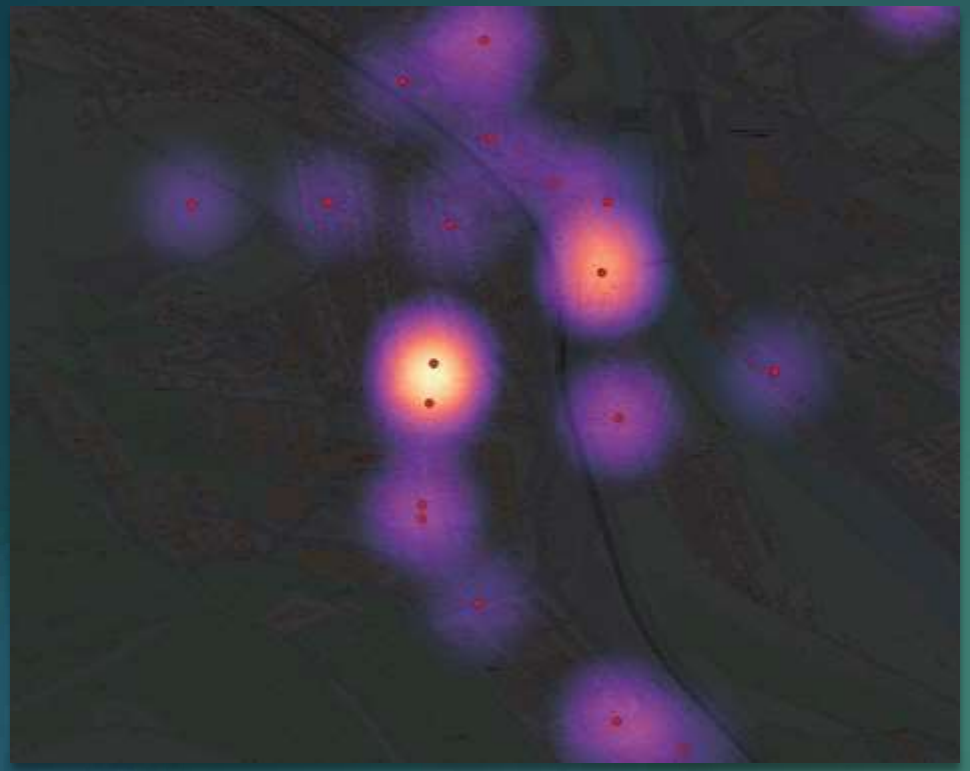
Heat Map for all crime in Jan 2018

This map represents all crimes that occurred within the Jan 2018 time period. As more than one crime may occur at each point, the radiating effect represents the percentage of total crimes that occurred at that point. The brighter the glow, the higher percentage of crimes.

Steps taken to produce map 2

To create this method, I followed the same steps used in map 1. However, to generate the heatmap effect with pointers overlaying I copied the Jan 2018 layer and changed the styling. The styling change is from single symbol to heatmap then selecting my desired colour gradient.





Comparison between Jan 2018 all crimes, and Jan 2019 all crimes

The map on the left shows all crimes reported in Jan 2018, while the map on the right shows all crimes reported in Jan 2019. These maps shows that the reported crimes have drifted northernly towards the town centre.

Steps taken to produce map 3

The first map used for the comparison is identical to the map used as map 2.

However, for the second map (Jan 2019) the step in creation are almost identical to that of the first comparison map, however when adding the crime data instead of adding Jan 2018, I add Jan 2019 crime data as the delimited text layer.

All violent crimes in Jan 2018

This map shows 4 spots where violent crimes were reported in Jan 2018. They are represented by a heatmap with the brighter colour representing a higher increase in reported violent crime.

Steps taken to produce map 4

This map uses a similar process to map two which generated a heat map with points. However to only show the relevant violent and sexual offences the following query was needed to be added to the crime data set

```
"Crime type" = 'Violence and sexual offences'
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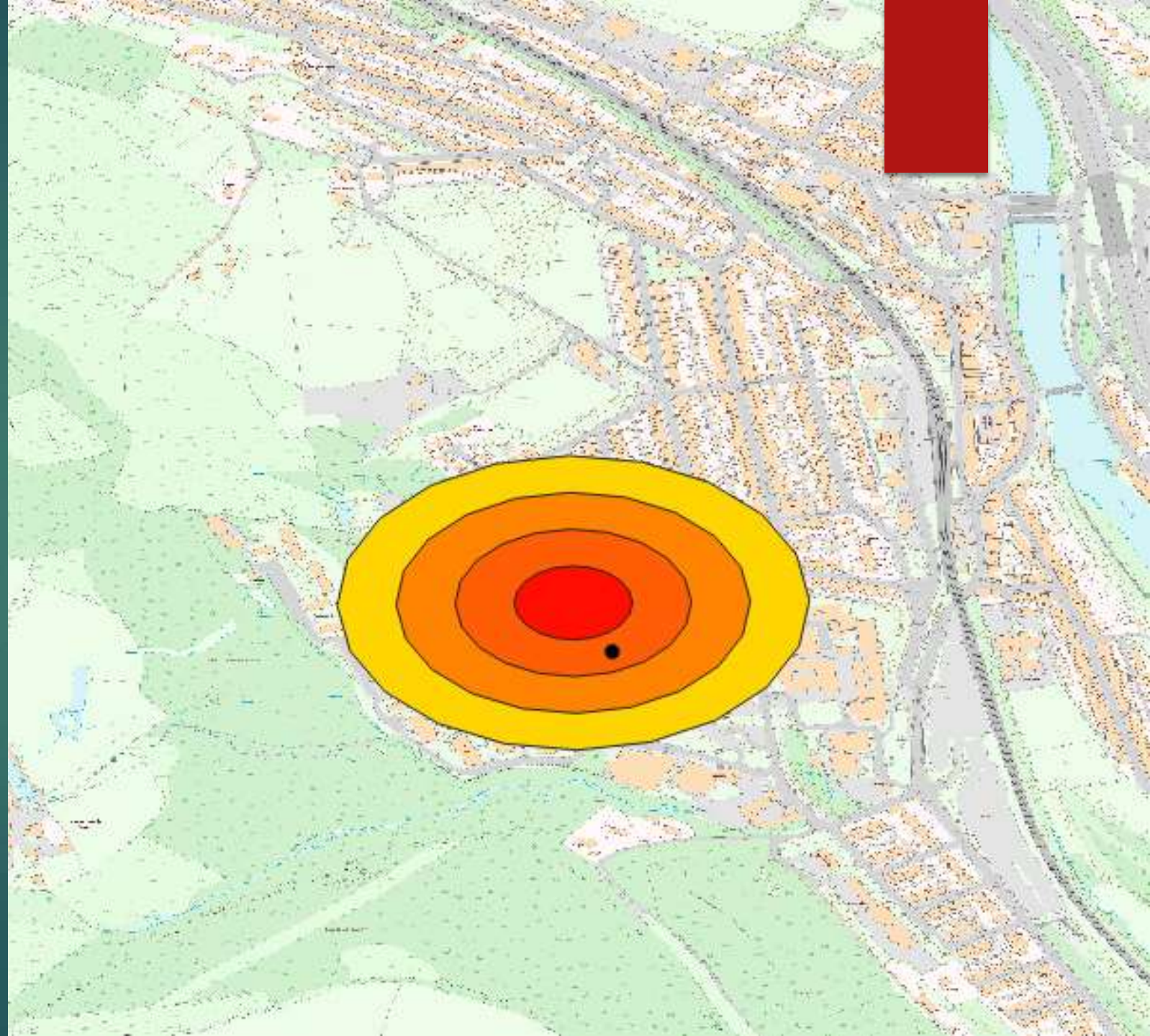
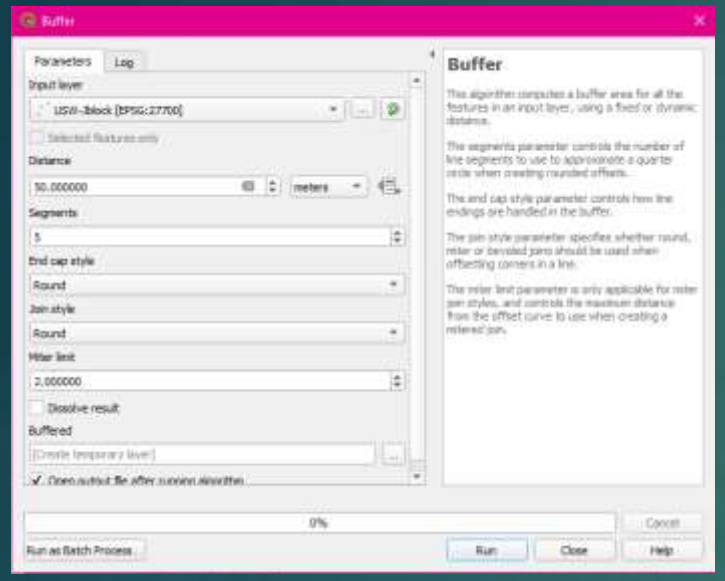
All Jan 2019 crimes that occurred within 200 meters of J block

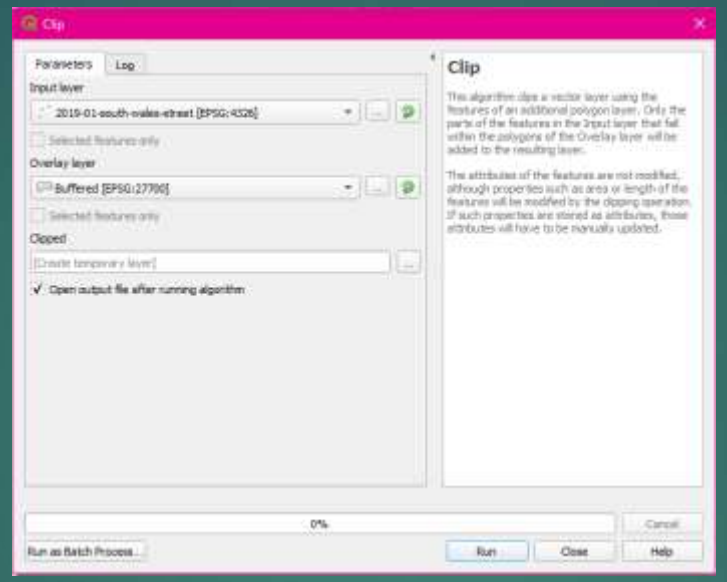
This buffer extents 200 meters radially from J Block, University of South wales. The buffer zone is spaced into colour gradients with increments of 50 meters.

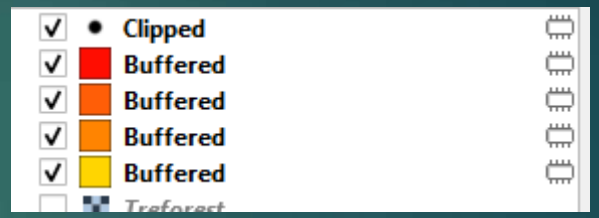
Only one crime happened with the 200 meters of J block, this occurred in the 2nd (100 meter) increment



Steps taken to produce map 5







The first step to having this spatial effect on the chosen map is to chose a point, that I've called USW_Jblock. Then adding a buffer around this point. Starting at 50 meters, I create another 3 buffer zones with 50 meter increments.

The second step is to clip all crimes to the largest buffer zone. This means only crimes that happened within the buffer zone (200 meters) will be displayed.

The final step is to order all the layers for them to display in the correct manner.

References

- Police.uk. (2019). *Treforest, South Wales Police - Police.uk*. [online] Available at: https://www.police.uk/south-wales/W05000699/ [Accessed 28 Mar. 2019].
- Digimap.edina.ac.uk. (2019). *Digimap*. [online] Available at: https://digimap.edina.ac.uk/ [Accessed 28 Mar. 2019].